

GEORGE BROWN POLYTECHNIC

Applied A.I. Solutions Development

PREDICTING EMAIL REPLY LIKELIHOOD

An NLP and Deep Learning Approach Using Azure Machine Learning

FINAL PROJECT REPORT

Business-focused NLP classification project designed to predict whether an outbound email is likely to receive a reply.

PROJECT INFORMATION

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Date	September 2026

PROJECT PROFILE

Project Domain	Natural Language Processing (NLP)
Business Task	Email Reply Likelihood Prediction
Dataset	Enron Email Corpus
Original Dataset Size	517,401 emails
Project Sample	20,000 emails
Manual Validation	300 emails
Training / Validation / Test	14,000 / 3,000 / 3,000 emails
Baseline Model	LSTM
Final Model	DistilBERT
Cloud Platform	Microsoft Azure Machine Learning

PROJECT WORKFLOW

Enron Corpus → Data Preparation → Reply Labeling → Manual Validation
LSTM → DistilBERT → Model Comparison → Azure Deployment

TOOLS & TECHNOLOGIES

Python • TensorFlow/Keras • PyTorch • Hugging Face • Azure Machine Learning

1. Introduction

1.1 Project Overview

This project aims to develop a Natural Language Processing (NLP) and deep learning system to predict whether an email is likely to receive a reply. The project uses the Enron Email Corpus as the source dataset and Microsoft Azure Machine Learning as the cloud environment for data processing, model development, and experimentation.

The project will compare a baseline **LSTM model** with a fine-tuned **DistilBERT model** for email reply likelihood prediction.

1.2 Business Problem

Business professionals often send many emails for communication, outreach, and follow-up. Predicting which emails are more likely to receive a response can help identify communication patterns and support better follow-up decisions.

The main question addressed by this project is:

Can the content of an outbound email be used to predict whether it is likely to receive a reply?

2. Azure Machine Learning Environment

2.1 Azure ML Workspace

An Azure Machine Learning workspace was created as the main cloud environment for the project. It provides a centralized location for managing data, notebooks, compute resources, experiments, and trained machine learning models.

Figure 1. Azure Machine Learning workspace successfully created.

2.2 Compute Instance

A CPU-based compute instance was created to provide the cloud computing resources required to run the project notebook, process the dataset, and develop the machine learning models. Automatic idle shutdown was configured after 30 minutes of inactivity to reduce unnecessary cloud computing costs.

Figure 2. Azure Machine Learning compute instance successfully created and running.

2.3 Project Notebook and Environment

A Jupyter Notebook was created within Azure Machine Learning and connected to the compute instance. A Python 3.10 environment with PyTorch and TensorFlow support was selected for data processing and deep learning development.

Figure 3. Project Jupyter Notebook created in Azure Machine Learning.

Figure 4. Azure compute and Python environment connected to the project notebook.

3. Data Understanding and Preparation

3.1 Dataset Source

The project uses the **Enron Email Corpus**, accessed through Hugging Face. The dataset was loaded directly into the Azure Machine Learning notebook using the Hugging Face datasets library.

The original dataset contains **517,401 email records and 9 fields**: message_id, subject, from, to, cc, bcc, date, body, and file_name.

3.2 Initial Data Inspection

Before sampling or cleaning the dataset, an initial inspection was performed to understand the structure and quality of the raw email data. Sample email records were reviewed to confirm that important information such as the subject, sender, recipient, date, and email body had been loaded correctly.

Table 1. Initial Dataset Inspection Results

Data Quality Check	Result
Total emails	517,401
Total fields	9
Missing/empty subject	19,187
Missing/empty sender	2
Shortest email	1 word
Longest email	60,526 words
Average email length	259.2 words
Date range	1980-2044

The initial inspection identified missing values, substantial variation in email length, and potential date anomalies that will be addressed during the data preparation and cleaning stage.

3.3 Create the Eligible Email Pool

After the initial inspection, the dataset was filtered to remove emails with unusual dates. Emails from **1997 to 2002** were kept because most of the Enron emails were found within this period.

After filtering, **516,796 emails** remained in the eligible email pool, while **605 emails with unusual dates were excluded**. The original dataset was kept unchanged for later reference and reply matching.

Table 2. Date Filtering Results

Data Preparation	Number of Emails
Original dataset	517,401
Eligible email pool	516,796
Emails excluded due to unusual dates	605

3.4 Remove Emails with Missing Required Fields

Emails with missing or empty subject or sender information were removed because these fields are required for the reply-matching process. A total of **19,117 emails were removed**, leaving **497,679 emails** for the next stage of data preparation.

Cleaning Result	Number of Emails
Before removing missing fields	516,796
Emails removed	19,117
Emails remaining	497,679

3.5 Create the 20,000-Email Project Sample

After the basic cleaning steps, **497,679 emails** remained. From this data, **20,000 emails were randomly selected** for the project. A fixed random seed of 42 was used to make the sample reproducible.

The 20,000-email sample will be used for reply labeling and model development, while the larger dataset will be kept for finding possible replies.

4. Email Reply Detection and Labeling

4.1 Reply detection Method

After creating the 20,000-email sample, each email was checked to determine whether it received a reply. The larger email pool of **497,679 emails** was used as a reference to search for possible replies.

Before matching, email subjects were normalized so that different versions of the same subject, such as **“Project Meeting”** and **“Re: Project Meeting,”** could be treated as the same subject.

An email was identified as receiving a reply when all of the following conditions were met:

- The original email and possible reply had the **same normalized subject**.
- The **sender and recipient were reversed**. For example, if the original email was sent from A to B, the reply had to be sent from B to A.
- The reply was sent **after the original email**.
- The reply was sent within **30 days** of the original email.

If all conditions were met, the email was labeled **1 (Reply)**. If no matching reply was found, it was labeled **0 (No Reply)**.

4.2 Target Label Distribution

After applying the reply identification method to all 20,000 sampled emails, **2,402 emails (12.01%)** were identified as receiving a reply, while **17,598 emails (87.99%)** were identified as not receiving a reply.

Table 3. Target Label Distribution

Label	Number of Emails	Percentage
Reply (1)	2,402	12.01%

Label	Number of Emails	Percentage
No Reply (0)	17,598	87.99%
Total	20,000	100%

The results show that the dataset has more **No Reply** emails than **Reply** emails. This class imbalance will be considered during model training and evaluation.

4.3 Manual Validation of Target Labels

A balanced sample of **300 emails** (150 Reply and 150 No Reply) was manually reviewed using the reply-identification criteria described in Section 4.1. The manual labels were compared with the automatically generated labels.

All **300 labels matched**, resulting in **100% agreement, precision, recall, and F1-score**. The confusion matrix contained 150 correctly identified Reply and 150 correctly identified No Reply cases, with no disagreements. This indicates strong consistency within the validation sample, although it does not guarantee that all 20,000 automatically generated labels are error-free.

5. Data Preprocessing and Model Preparation

5.1 Text Preparation and Dataset Split

The email **subject and body were combined** to create the text input used for model training. The 20,000 labeled emails were then divided into **70% training (14,000)**, **15% validation (3,000)**, and **15% testing (3,000)** using stratified sampling. This preserved the original class distribution, with approximately **12% Reply emails in each dataset**.

6. LSTM Model

6.1 LSTM Training and Evaluation

The LSTM model was trained using the prepared email text. The model used a **20,000-word vocabulary**, **200-token sequence length**, **128-dimensional embedding**, and **64 LSTM units**. **Class weights** were used because the dataset had fewer Reply emails, and **early stopping** stopped the training after 5 epochs to help prevent overfitting.

On the test set, the LSTM achieved **80.63% accuracy**, **18.16% precision**, **17.50% recall**, and **17.82% F1-score**. The high accuracy was mainly influenced by the large number of No Reply emails, while the low F1-score showed that the model had difficulty identifying Reply emails.

7. DistilBERT Model

A pretrained **DistilBERT model** was fine-tuned to predict whether an email would receive a reply. The model used the same training data as the LSTM, with **14,000 training emails**. Emails were tokenized with the DistilBERT tokenizer using a maximum length of **128 tokens**. A **batch size of 8** and class weights were used to handle the imbalanced Reply and No Reply classes. The model was fine-tuned for **one epoch**, with a final average training loss of **0.6301**.

The fine-tuned model was evaluated using the **3,000 unseen test emails**. It achieved **79.83% accuracy**, **27.44% precision**, **41.39% recall**, and **33.00% F1-score**.

8. Model Comparison

The LSTM and DistilBERT models were compared using the same test dataset. The LSTM achieved **80.63% accuracy and 17.82% F1-score**, while DistilBERT achieved **79.83% accuracy and 33.00% F1-score**. DistilBERT also achieved higher precision (**27.44%**) and recall (**41.39%**). These results show that DistilBERT was more effective at identifying Reply emails, although the LSTM had slightly higher overall accuracy.

9. Final Model Preparation

After comparing the models, the fine-tuned **DistilBERT model** was prepared for deployment. The model was first tested using a new sample email to confirm that it could generate predictions. It predicted **No Reply** with a **46.50% Reply probability**.

The fine-tuned DistilBERT model was then saved and registered in **Azure Machine Learning** as **email-reply-distilbert, Version 1**. This prepared the model for deployment and use with new email data.

10. Prototype Model Deployment

The registered **DistilBERT model** was deployed as a prototype using an **Azure Machine Learning managed online endpoint**. The endpoint allows new email text to be sent to the deployed model and returns a real-time **Reply or No Reply prediction** with a probability score.

The deployed model was successfully tested using a **new email that was not part of the training dataset**. The model predicted **No Reply** with a **41.84% Reply probability**, confirming that the deployed model could process new email text through the Azure online endpoint.

11. Conclusion

This project developed a deep learning system to predict whether an email is likely to receive a reply. The Enron Email Corpus was cleaned and prepared, and 20,000 emails were used for the project. The reply labels were also checked using a manual validation sample of 300 emails.

Two models were tested: LSTM and DistilBERT. The LSTM achieved 80.63% accuracy, while DistilBERT achieved 79.83% accuracy. However, DistilBERT performed better at finding Reply emails and achieved a higher F1-score of 33.00%, compared with 17.82% for LSTM. Final_Report

DistilBERT was selected as the final model and successfully deployed using Azure Machine Learning. The deployed model was able to take new email text and return a Reply or No Reply prediction. Overall, the project successfully completed the full process from data preparation and model training to testing and cloud deployment.